Anuj Pathak

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EDUCATION

Dr. D. Y. Patil Institute of Technology 2023 - 2027

B.E. in Mechanical Engineering, Honours in Artificial Intelligence & Machine Learning

Indian Institute of Technology Mandi 02/2025 - 05/2026

Minor Program, Artificial Intelligence & Data Science

EXPERIENCE

AI/ML Intern | Alfa Laval, Global Automation Development 05/2026 - Present

Pune, India

- Built and deployed reinforcement learning models to optimize decisions within business-critical automation workflows.
- Developed Retrieval-Augmented Generation (RAG) pipelines and agentic AI tools, integrating locally hosted and API-based LLMs into internal automation systems.
- Fine-tuned large language models using LoRA and QLoRA to improve task-specific performance under constrained compute.
- Collaborated in a cross-functional engineering team following Azure DevOps standards, including user stories, task creation and sprint-based execution.

PROJECTS

ThermoTrace: Inverse Heat Source Identification

Python, PyTorch, ANSYS (PyMAPDL), U-Net

- Designed a two-stage CNN regression and U-Net pipeline to recover internal heat-source locations and intensities from surface temperature distributions on a 2D steel plate.
- Automated thermal simulation data generation with PyMAPDL, producing labeled datasets and stable, low-error convergence across randomized configurations.

F1 Race Strategy Optimization with Reinforcement Learning

Python, PyTorch

- Modeled race strategy as a sequential decision problem and trained a DQN agent over thousands of simulations to learn pit-stop timing and tyre selection.
- Outperformed rule-based strategies by approximately 15% in simulated race finish position.

Sentiment Analysis System Using BERT

Python, Hugging Face Transformers

- Fine-tuned BERT on labeled reviews, reaching approximately 89% validation accuracy, with precision, recall and F1-score evaluation across positive, negative and neutral classes.

TECHNICAL SKILLS

Languages: Python
ML/AI: Reinforcement Learning, LLM Fine-tuning (LoRA, QLoRA), RAG, Agentic AI, CNNs, Computer Vision, NLP
Tools: PyTorch, Hugging Face Transformers, ANSYS (PyMAPDL), Azure DevOps, Git
Certifications: Supervised Machine Learning (Stanford); Career Essentials in Generative AI (Microsoft)

LEADERSHIP & ACTIVITIES

Technical Lead, Team Envisioners 04/2025 - Present

Lead and mentor 35 engineers and developers, driving technical architecture and team development.

AI/ML Member Lead, GDG (DIT Pimpri Chapter) 09/2025 - Present

Mentored students through hands-on ML modules during Google Cloud Study Jams.